



DESIGN YOUR OWN CIRCUIT!

The Grand Capstone Engineering Challenge

Grade 3 Capstone Project Guide

★ You are the Inventor today!
Combine batteries, lights, switches, and alarms
to make your very own working gadget!





Welcome, Junior Circuit Inventor!

Think about your favorite gadget at home. Is it a flashlight? A toy robot? Or a musical doorbell? 

All of those amazing toys have one big thing in common: inside them, hidden from view, is a ****working circuit****. A circuit is simply a complete, circular path that lets electric power travel around to complete chores for us.

Up until today, you have followed preset maps to build circuits. But today is your ****Grand Capstone Day****! That means the training wheels are coming off. You get to use your own imagination to sketch, plan, and build a unique gadget from scratch.

Are you ready to bring your own electronic idea to life? Let's take a look at the secret building blocks you can use!



Brainstorming Minute:

Close your eyes for ten seconds. If you could build any magical circuit machine right now to help you in your bedroom, what would it do? Turn and tell a classmate your idea!



Building Block 1: The Power Source

Every single electronic machine needs a starting push. Without it, electricity stays perfectly still and asleep inside the wires.

We use a **Battery** as our ultimate power pump! Think of a battery like a friendly heart that pumps electric current out into the circuit line.

-  **The Plus Side (+):** This is the high-energy exit gate. Electricity comes rushing out from here, excited to start its journey around your project table.
-  **The Minus Side (-):** This is the return gate. After electricity travels through your lights and buzzers, it circles back here to rest and recharge.



Remember the Rule:

Electricity is lazy! It will only leave the battery if it sees a clear, unbroken loop of wire leading all the way back to the minus side. If there's a gap, it won't move at all!



Building Block 2: The Action Outputs

Once your electric power is moving along the wire, you need to give it a fun job to perform! In engineering, components that do actions are called ****Outputs****.

You can choose from these fun action outputs in your kit box today:

💡 **The Bright LED Light:**

It turns electricity into a beautiful glow. You can use it to create a lighthouse beam, a toy car headlight, or a secret signal lamp! Remember to match its long leg to the plus wire.

🐝 **The Loud Piezo Buzzer:**

This component turns electric energy into a buzzing vibration sound! It is perfect for building a morning alarm clock, a game-show button, or a safety warning siren.

Which action output fits your project best? You can even combine them to make a device that lights up and buzzes at the exact same time!



Building Block 3: The Path Controllers

If you connect your battery directly to a buzzer, it will buzz non-stop forever until the battery dies. That would be way too noisy! We need a way to control the action.

To control our circuits, we add ****Switches****. A switch acts like a bridge operator over a river:

-  ****Bridge Closed (ON):**** When you press the switch down, the metal parts touch. The bridge is down, the path is complete, and electricity flies across to power your output!
-  ****Bridge Open (OFF):**** When you let go or flip it up, a gap is made in the wire. The bridge is pulled up, electricity stops in its tracks, and your machine turns off.



You can use a Press-Button Switch for quick blasts of power, or a Slide Switch to leave your machine running continuously!



Step 1: Draw Your Secret Blueprint

Real engineers never build blindly without a map. Before you touch a single real wire, you must sketch a blueprint layout on your project sheet!

Grab your pencil and draw simple symbols for your circuit parts. Connect them with lines that represent your conducting wires:



Blueprint Checklist:

1. Draw your battery box first and label the (+) and (-) sides.
2. Add your switch bridge in the middle of the positive line.
3. Draw your LED light or buzzer at the end of the line.
4. Draw a long line connecting your output back to the (-) side of the battery.

Make sure your pencil line forms a complete, unbroken loop from the battery, through the switch, into the light, and back home. No open gaps allowed yet!



Step 2: Gather and Check Your Materials

Now that your pencil blueprint map is approved by your teacher, walk over to the supply bench to collect your real hardware components.

Handle each component gently like a true scientist. Let's do a quick pre-test check on your breadboard bench:

Component Check	What to Look For
 Battery Snap	Are the red and black insulation coatings clean without tears?
 LED Wire Legs	Is the longer positive leg easily visible? Make sure it isn't bent or broken.
 Clip Wires	Do the metal alligator clips pinch tightly onto the component leads?



Step 3: Wire It Together Step-by-Step

Time to assemble! Follow your blueprint line just like tracking a treasure map.

Follow these construction steps carefully with your team:

1.  Clip a jumper wire from the ****Positive Red terminal**** of your battery holder over to the incoming terminal of your switch.
2.  Connect a second wire from the outgoing terminal of your switch to the ****Long Leg (+)**** of your LED light bulb.
3.  Connect a final wire from the ****Short Leg (-)**** of your LED light directly back to the ****Negative Black terminal**** of your battery pack.

Take a step back and inspect your work. Does the physical circle matching your drawn blueprint map exactly? Excellent job!



Troubleshooting: Fixing Broken Paths

What happens if you click your switch to ON, but your LED light stays completely dark? Don't worry! That is exactly what real engineers call ****Troubleshooting****.

Let's play detective and fix the bug using these 3 checks:

- ****Check 1: The LED Backwards Bug****
Is your LED long leg pointing toward the black wire? If yes, it's backwards! Flip the LED around so the long leg connects to the red battery side.
- ****Check 2: The Loose Clip Leak****
Wiggle your alligator clips gently. Are they pinching bare metal? If a clip is biting onto the plastic coating instead of the metal wire, the electricity is blocked.
- **CRITICAL SAFETY CHECK: The Heat Test**
If your battery pack feels warm to touch, disconnect it immediately! A wire is bypassing your output and causing a short circuit. Ask an adult helper to inspect the lines.

Capstone Finale: Present Your Invention!

Congratulations! Your circuit is officially up, running, and doing real work! 🏆

Now comes the final fun phase of being an engineer: showing off your creation to the world. Give your device an awesome name, like *"The Midnight Safety Beacon"* or *"The Anti-Monster Desk Alarm"*.

Share Your Project:

Stand proud at your lab bench and explain your gadget to the class:

**"I built a circuit that uses a _____ switch to power up a _____
whenever a user presses the bridge down!"**



You are now an official Circuit Designer! Keep on inventing!